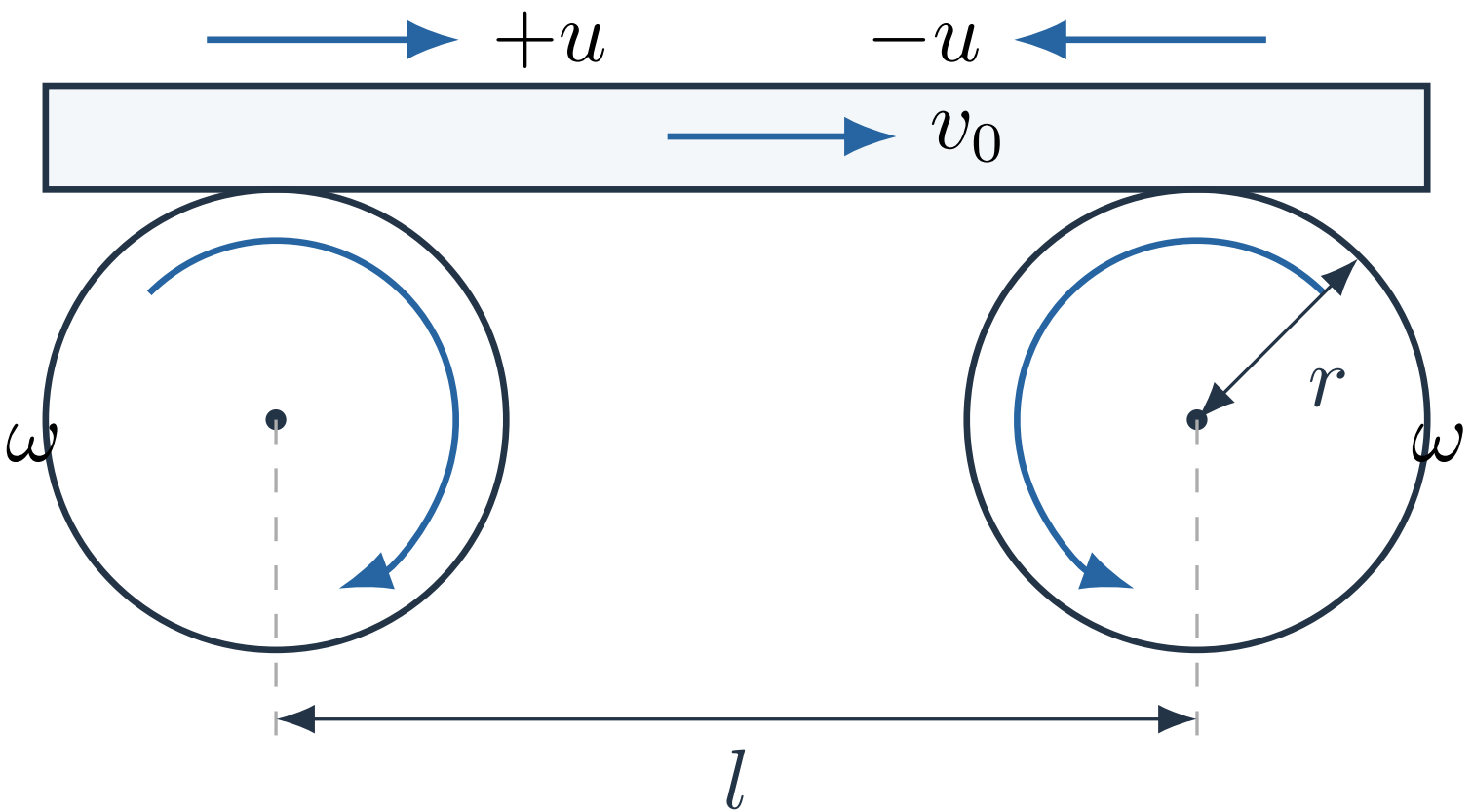
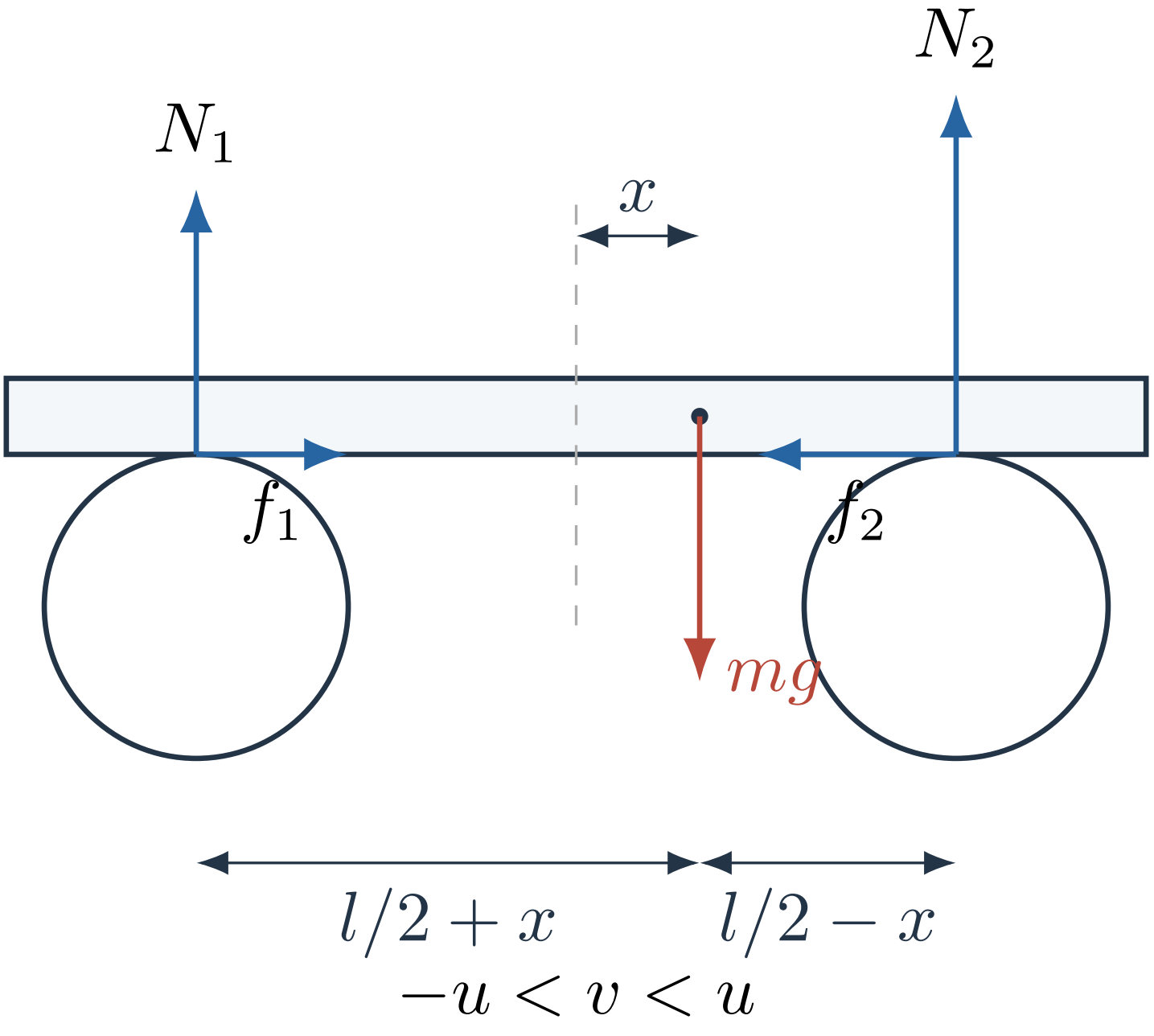
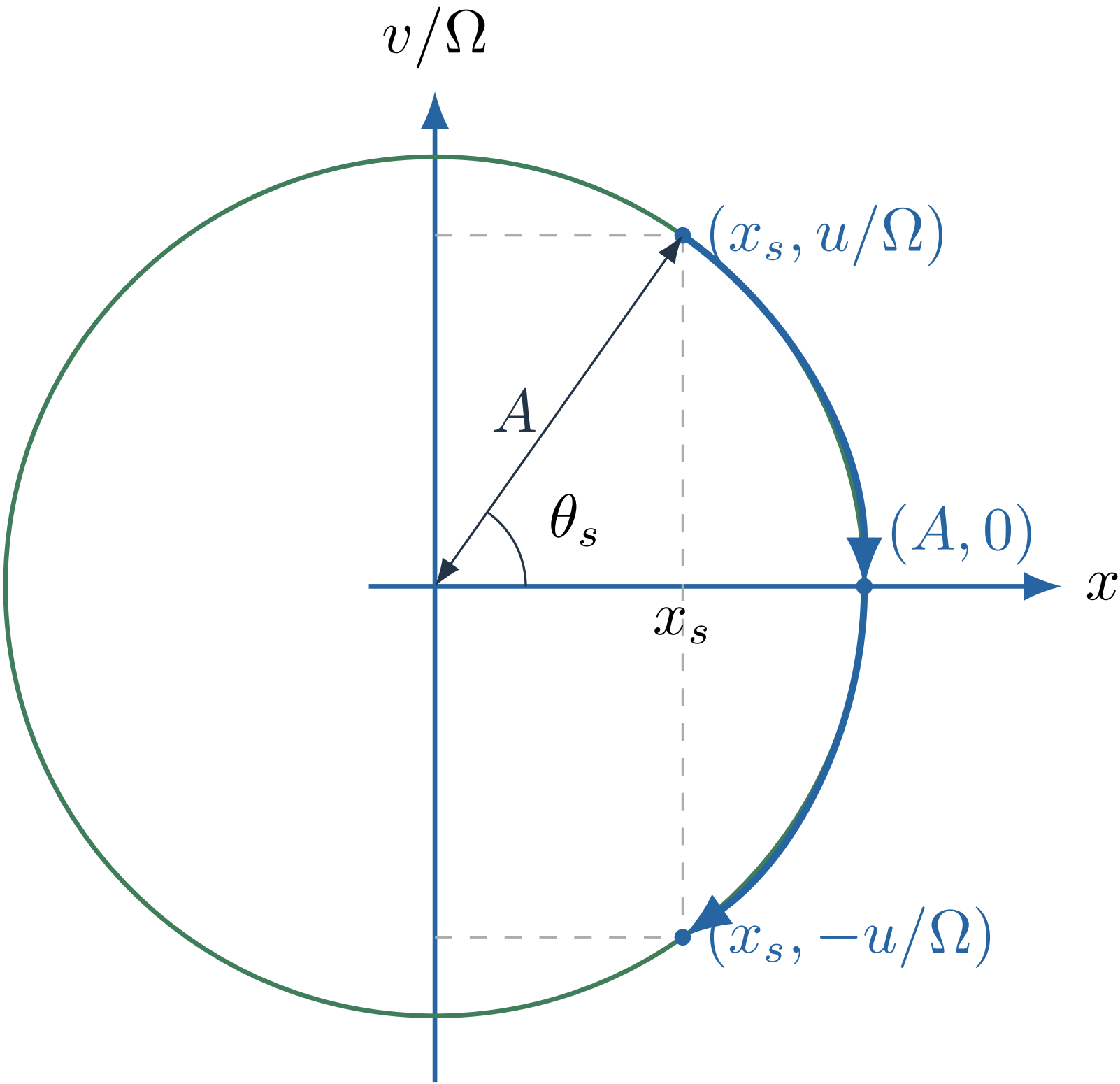

$+u$
$-u$
v_0
ω
ω
r
l

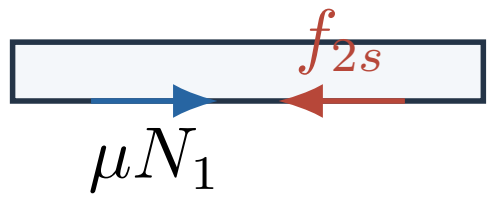

N_2
N_1
x
f_1
f_2
mg
$l/2 + x$
$l/2 - x$
$-u < v < u$


v/Ω
$(x_s, u/\Omega)$
A
θ_s
$(A, 0)$
x
x_s
$(x_s, -u/\Omega)$

简谐返程 $0 \to -u$

$|f_{2s}| = \mu N_1 \le \mu N_2$

简谐去程



匀减速 $v_0 \to u$

匀速 $v = -u$